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Ultra Water Repellent Polypropylene Surfaces with Tunable Water Adhesion

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Abstract:

Polypropylene (PP), including isotactic PP (i-PP) and atactic PP (a-PP) with distinct tacticity, is one of the most widely used general plastics. Herein, ultra water repellent PP coatings with tunable adhesion to water were prepared via a simple casting method. The pure i-PP coating shows a hierarchical morphology with micro-nano binary structures, exhibiting a water contact angle larger than 150° and a sliding angle less than 5° (for 5 μ L water droplet). In contrary, the pure a-PP coating has a less rough morphology, with a water contact angle of about 130° and the water droplets stick on the coating at any tilted angles. For the composite i-PP/a-PP coatings, however, ultra water repellency with $CA > 150^\circ$ but water adhesion tailorable from slippery to sticky can be realized, depending on the contents of a-PP and i-PP. The different wetting behaviors are due to the various microstructures of the composite coatings resulted from the distinct crystallization ability of a-PP and i-PP. Furthermore, the existence of a-PP in the composite coatings will enhance the mechanical properties compared to the i-PP coating. The proposed method is feasible to modify various substrates and potential applications in no-loss liquid transportation, slippery surfaces and patterned superhydrophobic surfaces are demonstrated.

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1. Introduction

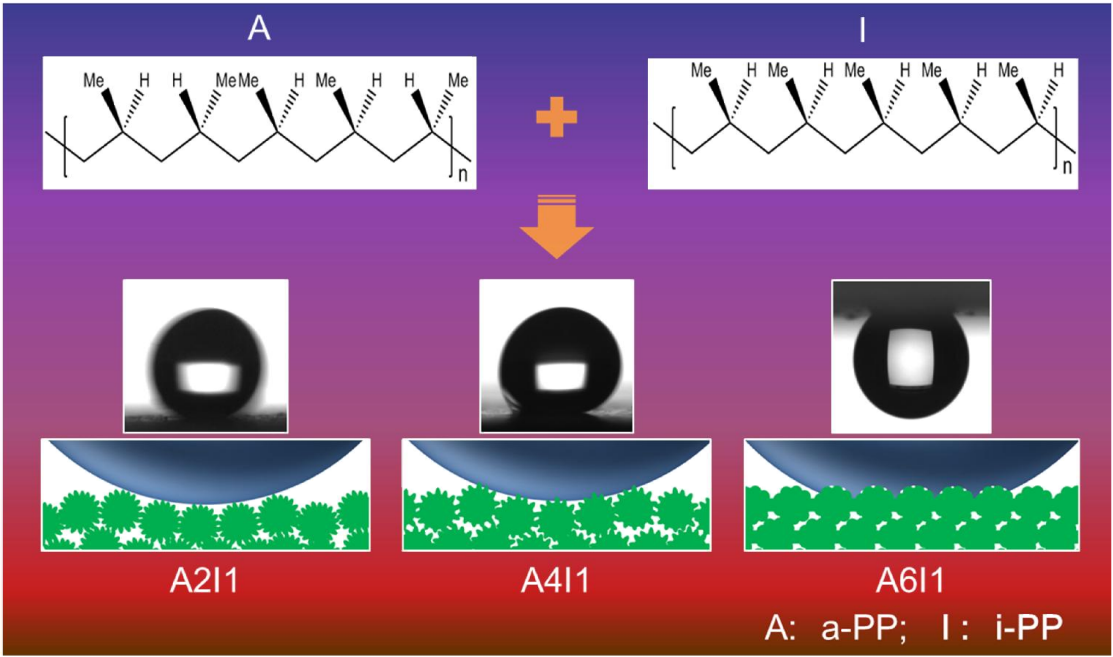
Ultra water repellent surfaces may show distinct water adhesion, and lotus leaf and rose petal are the typical examples. On the both surfaces, static water contact angles (CAs) are larger than 150°, however, water droplets can easily roll off the lotus leaf, while they stick on the rose petal at any tilted angles. The distinction can be mainly ascribed to their different surface microstructures, which results in different wetting states. To be specific, surface of lotus leaf is composed of micrometer-scale (5-9 μm in diameter) papillae decorated with nanometer-sized wax, and water droplet on which is in Cassie state as it cannot impregnate into the microstructures.¹ However, surface of rose petal is less rough, composed of micropapillae (about 16 μm in diameter) decorated with nanofold structures at the top. Water droplet on rose petal is in Cassie impregnating wetting state as water can impregnate between the microstructures and partially penetrate into the nanostructures.^{2,3}

Bio-inspired ultra water repellent surfaces with controllable water adhesion have attracted great interests due to their potential fundamental and industrial applications, such as in fluidic devices, liquid manipulation, micro-reactor, droplet-based bio-detector or bio-sensor.⁴⁻⁶ Surface morphology plays a critical role in determining the surface wettability, thus can be tailored to obtain tunable adhesion. Lai et al.⁷ proposed that surface adhesive force is closely related to solid-liquid contact means and the magnitude follows that “area-contact” is larger than “line-contact”, which is larger than “point-contact”. Bhushan³ systematically studied various rose petals and indicated that the surfaces with larger pitch value, smaller peak-to-base height or fewer nanostructures possess higher adhesion. Femtosecond laser irradiation has been used to control the microstructures on various substrates by tuning the scanning speed,

laser power or average distance of adjacent microstructures.^{5,8-14} Controlling crystalline condition to construct different microstructures is also a valid approach. For example, by adjusting the shear rates and thermal histories, different crystal morphologies of i-PP have been obtained.¹⁵ By changing the current densities of the galvanostatic electrochemical deposition, nano-crystals of cobalt with diverse morphologies can be prepared.¹⁶ Various crystals of photochromic diarylethene derivative have also been fabricated via controlling UV irradiation and storage temperature.¹⁷ Anodization, hydrothermal reaction, redox reaction were also induced to create surfaces from “no-sticky” to “sticky”. However, these methods are mainly restricted to inorganic materials, such as TiO_2 ¹⁸, Al_2O_3 ¹⁹, CuO ²⁰, $\text{Cu}(\text{OH})_2$ ²¹, MnO_2 ²², ZnO ²³. Other technologies like replica technology²⁴, plasma etching²⁵⁻²⁷ have also been utilized to create tunable microstructures. Although all these approaches can be successfully applied to prepare water repellent surfaces with controllable water adhesion, there are some disadvantages including the requirement of specific equipment, complicated and time-consuming processes, harsh fabricating conditions, high energy consumption or unavailability for large-scale production. It is still a challenge for facile and economic manufacture of bio-inspired ultra water repellent surfaces, especially using commercial polymers.

PP is the second-most widely produced synthetic plastic in the world. In this research, ultra water repellent PP coatings with tunable water dynamic behaviors from slippery to sticky were simply prepared by casting solutions containing different ratio of i-PP to a-PP. The distinct wetting behaviors can be ascribed to the diverse wetting states caused by the different microstructures, which are resulted from the distinct crystallization behaviors of i-PP and a-PP (Scheme 1). Moreover, the mechanical

properties of the coatings are enhanced by adding a-PP into i-PP. This simple method is feasible to modify various materials, and shows merits in the applications such as self-cleaning, no-loss liquid transportation, slippery surfaces and patterned superhydrophobic surfaces.



Scheme 1. Schematic illustration of the ultra water repellent PP coatings with tunable adhesion to water. The coatings have different microstructures depending on the ratio of a-PP to i-PP, and water on these coatings exhibits distinct wetting states, varying from Cassie state to Cassie impregnating wetting state.

2. Experimental Section

2.1 Materials

Granular i-PP ($M_w = 12$ kDa, $M_n = 5$ kDa, $d = 0.9$ g/mL at 25 °C) and a-PP ($M_w = 20$ kDa and $M_n = 5$ kDa, $d = 0.9$ g/mL at 25 °C) were purchased from Aldrich. Xylene,

cyclohexane, and hexadecane of analytical grade were obtained from Beijing Chemical Works. All chemicals were used as received without further purification.

2.2 Fabrication of PP Coatings

Briefly, 0.45 g i-PP or a-PP was dissolved in 30 mL xylene in a 50 mL flask sitting in a silicone oil bath at 120 °C. The clear solution was cast on the clean substrates by a dropper under ambient atmosphere, and the amount of the casting solution is about 0.15 mL/cm². Glass slide (76 mm × 26 mm), PTFE plate (70 mm × 25 mm) and Al sheet (75 mm × 35 mm) are coated. After the evaporation of the solvent, i-PP or a-PP coating was obtained. Similarly, the composite coatings of a-PP and i-PP with different mass ratios (2/1, 4/1 and 6/1) were prepared by the same method. In addition, smooth i-PP and a-PP films were also prepared by heating the as-prepared i-PP and a-PP coatings at 200°C and then cooling to the room temperature rapidly.

2.3 Characterization

2.3.1 Characterization of static contact angles and dynamic contact angles

Static water CA, advancing CA (θ_{adv}) and receding CA (θ_{rec}) were measured using a drop shape analysis instrument (KRÜSS DSA 100, Germany) at ambient temperature via a sessile drop method. A water droplet of 5 μ L was employed and each contact angle value was an average of five measurements on different positions of the coating. Advancing or receding CA was measured by sequentially increasing or decreasing the volume of the water droplet deposited on the surface with a small syringe and was recorded as the contact line begin to advance or retract, and the CA hysteresis was defined as ($\theta_{adv}-\theta_{rec}$). Sliding angles (SAs) were also measured five times for each sample by tilting the coating until the droplet (5 μ L) began to slide.

2.3.2 Characterization of surface adhesion to water

The adhesion force was characterized by a high-sensitivity micro-electromechanical balance system (Dataphysics DCAT 11, Germany). A water droplet (15 μL) was suspended on a metal ring which was fixed to the balance, and the coating was placed on the balance stage under the metal ring. The water droplet was firstly moved downward to the coating at a constant speed of 2 mm/s until the droplet contacted the coating surface. Then the droplet continued to immerse 0.5 mm to the surface with the motor speed of 0.2 mm/s. After that, the water droplet was pulled up slowly to break away from the coating. During the last process, the water droplet was stretched from the spherical shape to elliptical one, and the force increased gradually to the maximum. When the droplet broke away from the surface, it changed back to the spherical shape and the force rapidly reduced down to zero. The maximum force recorded at the point when the droplet was just leaving the surface was defined as the adhesive force. Work of adhesion was then obtained by integrating the areas of adhesion force and distance. The value of adhesive force and the corresponding adhesive work of each coating were an average of five measurements made at different locations.

2.3.3 Characterization of water evaporation process

The same instrument (KRÜSS DSA 100, Germany) was used to observe the change of water droplet over time due to evaporation. Photos were taken every two minutes during the evaporation process until the droplet evaporated completely. The CA, contact diameter (CD) and droplet height (H) were recorded and analyzed. Several droplets were observed during their evaporation on each sample, showing

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3 good statistical consistency. A typical evaporation curve of each sample was
4 presented here.
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7 8 *2.3.4 Characterization of the mechanical properties of the coating* 9

10 The mechanical properties of the coating were characterized by cross-cut test
11 according to ISO 2409-2013. All the tests were carried out at a temperature of $23 \pm$
12 $2\text{ }^{\circ}\text{C}$ and a relative humidity of $50 \pm 5\%$. A cutter with a spacing edge of 1 mm was
13 selected since the thickness of the coating is 20-30 μm (less than 60 μm). Firstly,
14 create a cross-cut lattice pattern: several parallel cuts to expose the substrate were
15 made with the cutter blade perpendicular to the coating at a uniform pressure and
16 cutting rate, and the operation was repeated to make further several parallel cuts,
17 crossing the original cuts at 90° . Then remove any loose powder from the area of
18 cutting by brush gently. After that, the tape SG-3M (600-1PK) was applied on the
19 pattern and then it was torn off at a small angle. Three different places were tested.
20 The adhesion of the coatings to substrate was valuated according to the area
21 percentage of peeling off.
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38 39 *2.3.5 Characterization of morphologies and crystallization behaviors* 40

41 The formation of the coating was observed by optical microscope (OM)
42 (Olympus BX-51, Japan) and the surface morphology was investigated by scanning
43 electron microscope (SEM, JEOL-7500, Japan) operated at 5 kV. Differential
44 scanning calorimetry (DSC) measurements were performed using DSC Q2000 TA
45 Instruments-Waters LLC (America) at a heating/cooling rate of $10\text{ }^{\circ}\text{C}/\text{min}$ in nitrogen
46 atmosphere. Wide angle X-ray diffraction (WXR) was performed on P2
47 X-ray polycrystal diffractometer (PANalytical, Holland). Cu was used as the anode
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material, the generator voltage was 40 kV, the tube current was 40 mA, and the scan angle (2θ) ranges from 5° to 30° .

3. Results and Discussion

3.1 Morphologies

Figure 1a shows the typical top-view SEM image of the obtained i-PP coating. The coating was constructed by the aggregation of nearly discrete sphere-like particles with the diameter ranging from 2 μm to 5 μm . From the magnified image, it can be seen that the surface of the particle is covered with smaller protuberances. In contrast, the a-PP coating prepared was not as rough as the i-PP coating. Figure 1b shows that the a-PP coating was built up by the coalescence of smaller particles. These a-PP particles immersed to each other without clear outlines and more importantly, the surface of the a-PP particles are less rough compared to that of i-PP particles (Insets in Figure 1a and b). For the composite coatings A2I1, A4I1 and A6I1, of which the mass ratio of a-PP to i-PP is 2/1, 4/1 and 6/1, respectively, the corresponding SEM images are shown in Figure 1c-e. All the three coatings are constructed by the aggregation of particles, however, the surface roughness decreases gradually with the increase of the content of a-PP. Noticeably, the outline of each sphere in the composite coatings becomes more obscure as the content of a-PP increases.

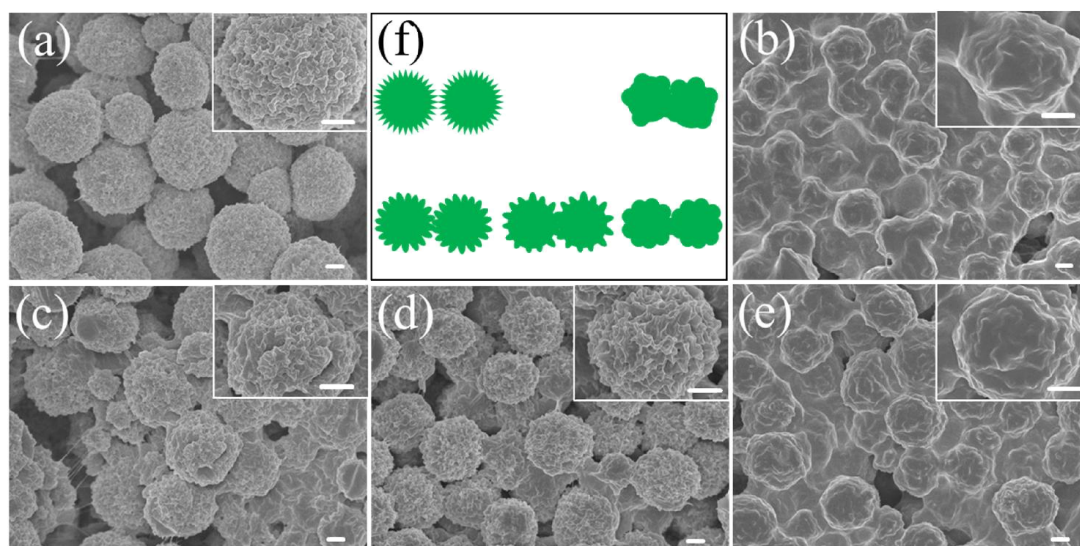


Figure 1. Top-view SEM images of i-PP (a), a-PP (b), A2I1(c), A4I1 (d), A6I1 (e) coatings. The corresponding magnified images are shown in the upper right corner on each panel. Scale bars: 1 μm . The insert (f) between (a) and (b) is the schematic illustration of the typical building blocks of the as-prepared PP coatings.

The different morphologies of PP coatings are mainly due to the different crystallization behaviors of PP chains with distinct regularity. While i-PP, a-PP and the composite coatings have the same crystal forms (WXR D results in Figure S1 in supporting information), i-PP is more easily crystallized and has the highest crystallization degree, and the crystallization degree of the composite coatings decrease with the increase of a-PP content (DSC results in Figure S2 and Table S1 in supporting information). Optical microscope was utilized to investigate the formation of the coating (Movie S1-3 in supporting information). The crystal nucleus is too small to be observed during the initial stage, however, i-PP coating turned to be opaque firstly, followed by A4I1 composite coating and then a-PP coating. This is because i-PP with a better regularity shows a poor solubility and rapid crystallization rate compared to a-PP. Therefore, phase separation is more easily occurred in i-PP

solution during the casting process, thus the i-PP coating turned to be opaque more quickly than the other solutions. For the composite coating, the “hard” i-PP chains play the dominant role in the formation of crystal nuclei and the “soft” a-PP chains are more likely to enclose the nuclei later. This speculation was confirmed by the etching experiment. Because a-PP can be dissolved in cyclohexane at 60 °C while i-PP cannot, cyclohexane was chosen to selectively etch a-PP in the composite coating.²⁸ After 8 hours etching, isolated smaller particles with rough surfaces remain on the substrate (Figure S3 in supporting information). The results revealed that in the composite coating, a-PP acted as the continuous phase trends to enclose the dispersed i-PP phase. This is in agree with the SEM images, from which it can be seen that for the composite coatings the particles are contact with each other more closely with the increase of a-PP concentration in the coatings.

3.2 Static and Dynamic Wetting Behaviors

Static Water CA of i-PP coating is $160 \pm 3^\circ$ and the SA is smaller than 5° , while water CA of a-PP surface is only $135 \pm 2^\circ$ and the droplet pins on the coating and doesnot slide even at a tilted angle of 90° (Figure 2). Since the surface wettability is governed by the surface tension and the topmost microstructures,²⁹ and the surface tension of i-PP and a-PP are similar (29.4 dyn/cm^{30}). In addition, the static CA of the smooth i-PP film is $103 \pm 2^\circ$ with a CA hysteresis of 34° , while the static CA and CA hysteresis of the smooth a-PP film are $102 \pm 1^\circ$ and 35° , respectively (Figure S4 in supporting information)). Thus it can be concluded that the difference of the wettability between i-PP and a-PP coating can be ascribed to their different microstructures. The CAs of the composite coatings A2I1, A4I1 and A6I1 are $159 \pm 4^\circ$, $154 \pm 3^\circ$ and $152 \pm 3^\circ$, respectively (Figure 2). Water CA slightly decreases with the

content of a-PP, however, the dynamic wetting behaviors of the composite coatings are rather different. Water droplet on A2I1 is easy to slide off with a SA smaller than 5° , and the SA of A4I1 is $30 \pm 7^\circ$. In contrast, water droplet pins on A6I1 and does not slide even the coating tilted at 90° (Figure 2). Further, the advancing CA (θ_{adv}), receding CA (θ_{rec}) and CA hysteresis ($\theta_{adv} - \theta_{rec}$) identified the different dynamic wetting behaviors. Figure S5 shows that the CA hysteresis is smaller than 5° for the i-PP and A2I1 coating, and is larger than 90° for the A6I1 and a-PP coating, and is 59° for the A4I1 coating. The difference can be ascribed to their different microstructures, which makes water droplet on the surface exhibit distinct wetting state. For A2I1, air trapped in the cavities of the rough surface hampers the penetration of the liquid, and the droplet on the surface displays Cassie model.^{31, 32} When the mass ratio of a-PP increases, the surface roughness decreases, and less air can be trapped in the microstructures. Therefore, water on A6I1 displays Cassie impregnating wetting state because water can impregnate between the microstructures but partially penetrate into the nanostructures, while water on A4I1 displays the transitional state between Cassie and Cassie impregnating wetting states (Scheme 1).^{2,33,34}

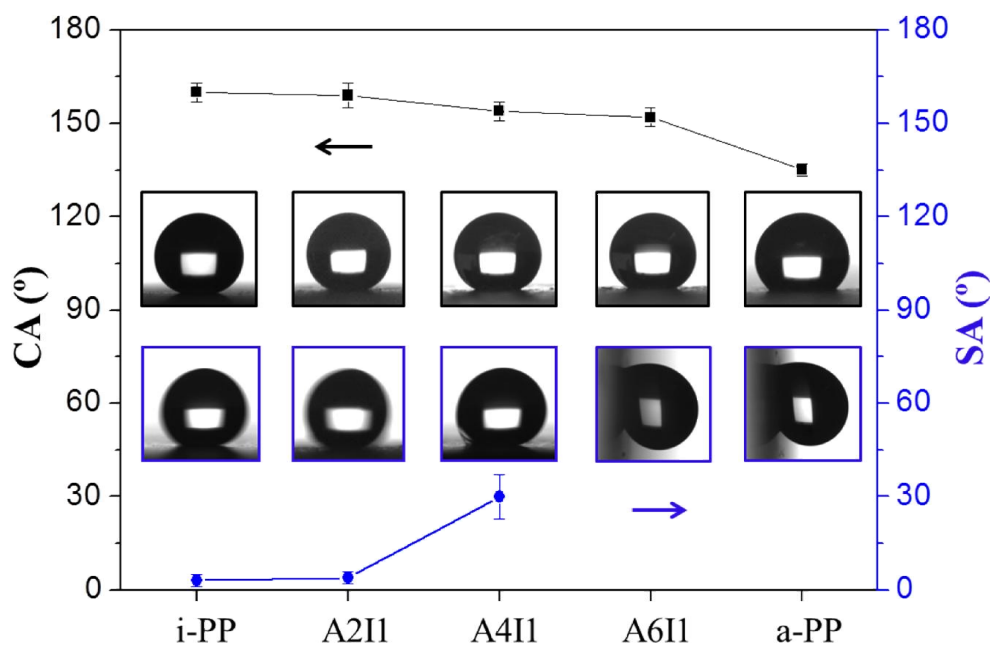


Figure 2. CAs and SAs of water droplet (5 uL) on different PP coatings. The inserts are the profiles of the corresponding water droplets showing the distinct wetting behaviors.

Furthermore, dynamic water adhesion measurements were performed. The test process contains three steps, which are shown in Figure 3a. Firstly, the water droplet was moved toward the surface until it contacted the surface. Then, the droplet continued to immerse to the surface for 0.5 mm, making the contact area between water and the surface increase gradually. Finally, the water droplet was pulled up from the surface. In this process, the water droplet was stretched to elliptical shape, and the force increased gradually to the maximum. When the droplet broke away from the surface, the shape returned to spherical and the force reduced to zero. The typical force-distance curves of these coatings indicate their distinct adhesion behaviors (Figure 3b). For a-PP coating, the water droplet cannot be pulled up entirely, and most

of the droplet was left on the surface, so the maximum force detected cannot represent the real adhesion force of the coating to water. The adhesion forces of water droplet to i-PP, A2I1, A4I1, and A6I1 are about 10.55 ± 1.83 , 15.24 ± 2.57 , 153.56 ± 25.20 and 188.25 ± 9.19 μN , respectively (Figure 3c), confirming that tunable water adhesion has been achieved by adjusting the mass ratio of a-PP and i-PP. The adhesion work calculated are about $(3.23 \pm 0.94) \times 10^{-9}$, $(3.9 \pm 1.3) \times 10^{-9}$, $(153.7 \pm 51.4) \times 10^{-9}$ and $(375.1 \pm 29.8) \times 10^{-9}$ J for coating i-PP, A2I1, A4I1 and A6I1, respectively. The adhesion work for A6I1 is about 90 times higher than that of A2I1, and the water adhesion force can be tuned ranging from 11 μN to 188 μN , which is comparative with the present reported data. In addition, the inserts in Figure 3b show the critical state of water droplet just being pulled away from the corresponding surface. The droplet on A2I1 keeps almost spherical while a large deformation was observed on A6I1. The deformation of water on coating A4I1 is between that on A2I1 and A6I1. Besides, the dynamic water adhesion behaviors have a certain relationship with the dynamic water CAs. For coating i-PP and A2I1 with a higher roughness, both the sliding angle and CA hysteresis are smaller than 5° , and the receding angles are 159° and 157° , respectively. That means the three phase contact lines on both of the coatings are easy to change, leading to a very low water adhesion of the surfaces. In contrast, the sliding angle and CA hysteresis of the less rough coatings A6I1 and a-PP are larger than 90° , and the receding angles are 45° and 20° , respectively. The three phase contact lines on these two coatings are preferred to pin on the surfaces, thus why a high water adhesion is observed. For the A4I1 coating with a moderate roughness, CA hysteresis and adhesion force between that of A2I1 and A6I1 coatings

are obtained. Thus, dynamic behaviors of water on the coatings can be modulated by the proportion of i-PP and a-PP.

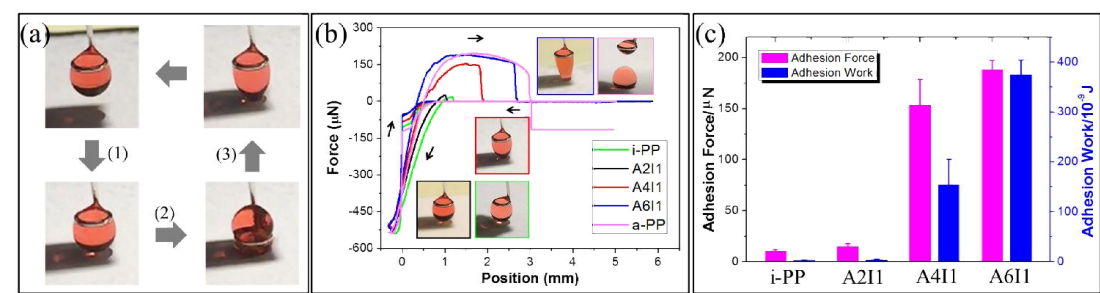


Figure 3. Dynamic adhesion measurement on the coatings: (a) photographs showing a typical measurement process. (b) Typical force-distance curves for the coatings. The inserts are the critical state of the droplet just leaving the corresponding surface. (c) The maximum adhesion force and adhesion work of the coatings.

The different evaporation means of water droplet on the coatings also give a proof of the diverse wetting states. The changes in height (H), CA and liquid-solid contact diameter (CD) of 5 μL water droplet during the evaporation on A2I1 with a low water adhesion and on A6I1 with a high water adhesion were compared. The height of water droplet decreases with time on both surfaces (Figure 4a-c). However, the decreasing rate for water on A6I1 is faster than that on A2I1. The CD of water droplet on high adhesion surface remains approximately constant for up to 40 min before decreasing sharply, whereas the CD on the low adhesion surface decreases gradually during evaporation (Figure 4a, b, d). For the CA of the water droplet, it decreases gradually on the surface with high adhesion while it keeps constant for nearly 40 min on the surface with low adhesion, after which it decreases sharply (Figure 4e). Therefore, the evaporation of water droplet on high adhesion surface obeys the constant CD mode, while it follows the constant CA mode on low adhesion

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3 surface.^{35, 36} The difference is mainly due to the different movements of the three
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6 phase contact lines on the two coatings: the small adhesion surface has a low contact
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8 angle hysteresis, thus the contact line is easy to change during the evaporation, so the
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10 contact area shrinks with the time, CA changes a little and H decreases slowly; on the
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12 contrary, the high adhesion surface shows a large contact angle hysteresis, and the
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14 contact line pins during the evaporation, so the contact area does not shrink, CA
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16 decreases gradually and H decreases more obviously.³⁷⁻³⁹ The different evaporation
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18 behaviors further confirm the water droplets on the composite coatings exhibit diverse
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20 wetting states due to the different surface microstructures.
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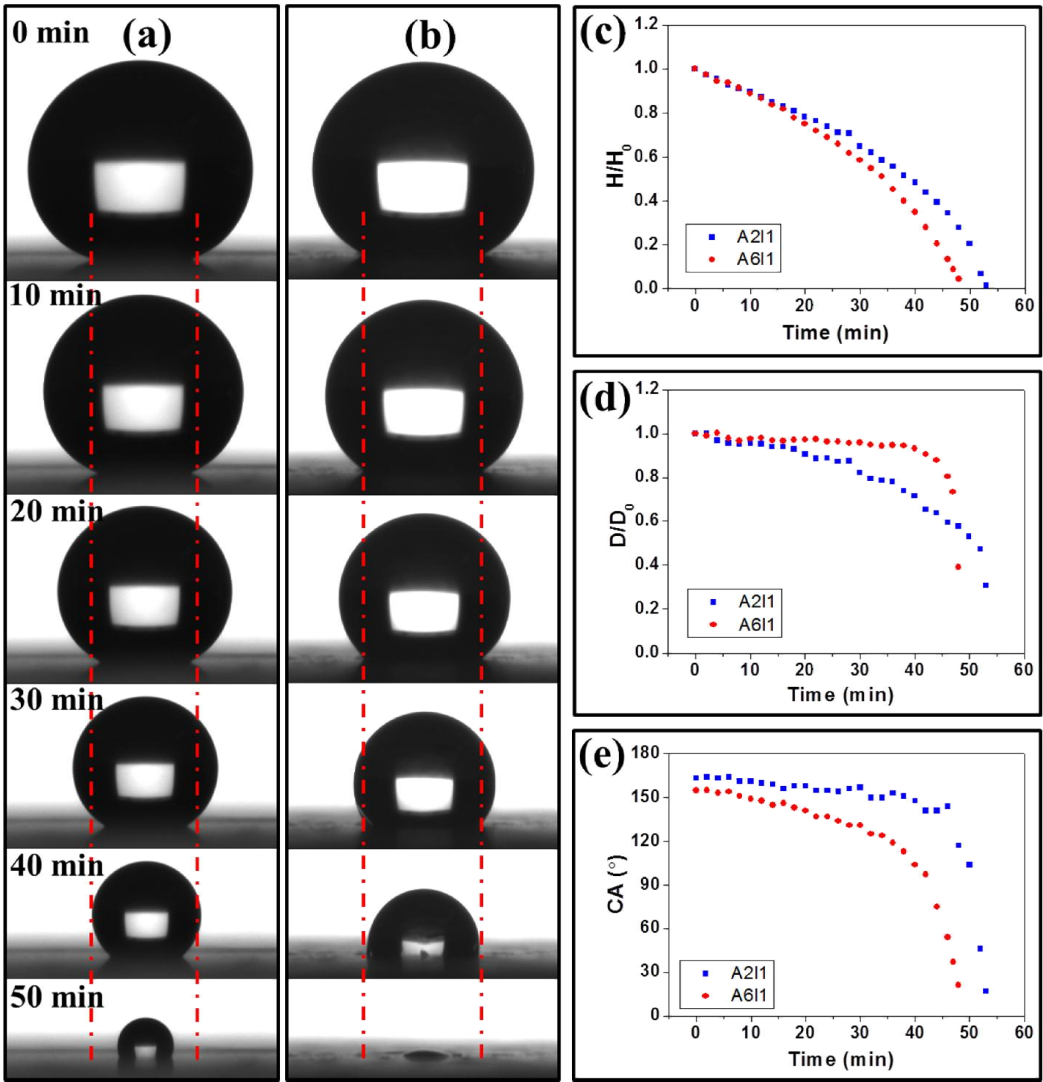


Figure 4. Real-time evolution of the water droplet during evaporation on water ultra repellent surface with low (a) or high (b) adhesion to water, and the CA image is collected every 10 minutes. Evolution of height (c), CD (d) and CA (e) for water droplets (5 μ L) evaporating on coatings with high (A6I1) or low (A2I1) adhesion. H and D are the height and contact diameter of the water droplet, respectively. H_0 and D_0 are the initial height and contact diameter of the droplet before evaporation.

3.3 Mechanical Properties

In fact, poor mechanical properties impede the practical applications of the ultra water repellent coatings. Here, the mechanical properties of composite PP coatings are enhanced due to the existence of a-PP. SEM images (Figure S6) show that cracks are enhanced due to the existence of a-PP. SEM images (Figure S6) show that cracks larger than 10 μm can be clearly seen in i-PP coating, however, almost no cracks are found in a-PP coating. For the composite coatings, the cracks in A2I1 are obviously smaller, and almost no cracks can be observed in A4I1 and A6I1 (Figure S6b-d). Further, the cross-sectional SEM images of various PP coatings are shown in Figure 5. There is no adjacent part between particles in the i-PP coating (Figure 5a) but the building blocks for a-PP coating are all adhered to each other (Figure 5e). For the composite coatings, however, the mutual contact of the building blocks tends to be stronger with the increase of a-PP (Figure 5b-d). The more interactions among the building blocks and also between the coating and the substrate lead to a good adhesion of the coating and a sound coating mechanical property.

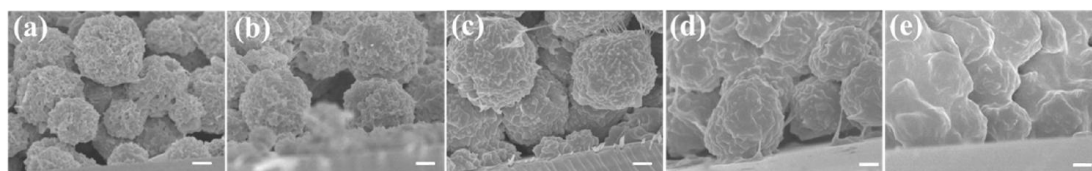


Figure 5. Cross-sectional SEM images of i-PP coating (a), i-PP and a-PP composite coatings of A2I1 (b), A4I1 (c), A6I1 (d), and a-PP coating (e). Scale bars: 1 μm .

The mechanical properties of the coatings are testified by cross-cut test according to ISO 2409-2013. As shown in Figure 6, the upper displayed at each subfigure is the lattice patterns created by the cutter, and the bottom shows the photo after the tape is applied. The coating flakes along the cutting edges and at the intersections when creating the patterns on i-PP coating, and the ratio of the flaked area ranges from 5% to 15%. More serious detachment can be observed after tape is applied: more cutting

edges and intersections are exfoliated, and some squares flake partly or wholly, and the flaked area ratio ranges from 35% to 65% (Figure 6a). However, only a few edges and intersections show detachment when creating the patterns on a-PP coating, and less than 5% cross-cut areas are affected. The area detachment increases slightly and remains in the range of 5% after the tape is torn off (Figure 6e). For the composite coatings, the flaked area ratio decreases as a-PP increases. Specifically, for A2I1 and A4I1 coating, the ratios of flaked areas before and after tape applied are in the range of 15% to 35%, which are smaller than that of i-PP (Figure 6b, c). And for A6I1 coating, the detached area ratio is smaller than 5% after creating the patterns, and only a few new edges and intersections appear after the tape is applied (Figure 6d). Moreover, the coating adhesions to PTFE plate and Aluminum sheet were further studied (Figure S7, 8). The i-PP coating shows the worst adhesion while the a-PP coating exhibits the best performance. For the composite coatings, the adhesion is enhanced with the amount of proportion of a-PP, no matter what kind of substrate is used. The above tests demonstrate that the mechanical properties of the composite coatings increase with the content of a-PP, no matter which substrate is used.

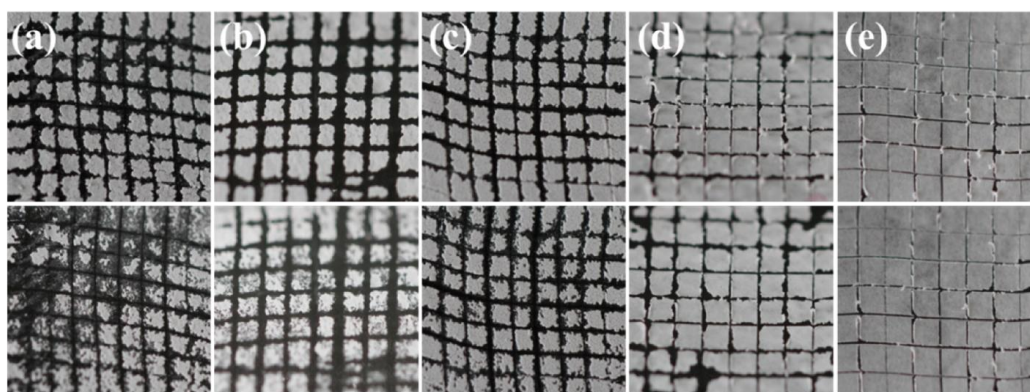


Figure 6. Digital photos showing the results of the cross-cut test: i-PP (a), A2I1 (b), A4I1 (c), A6I1 (d) and a-PP (e) coatings. The upper and the bottom photos in each subfigure are taken before and after tape applied, respectively.

3.4 Versatility

The proposed method is feasible to modify various materials, including inorganic, metal, polymer substrates and porous materials. For example, as shown in Figure 7, filter paper, stainless steel mesh and gauze become superhydrophobic by casting the A2I1 solution on these substrates, and the superhydrophobic PU foam is also obtained via a dip-coating method in A2I1 solution.

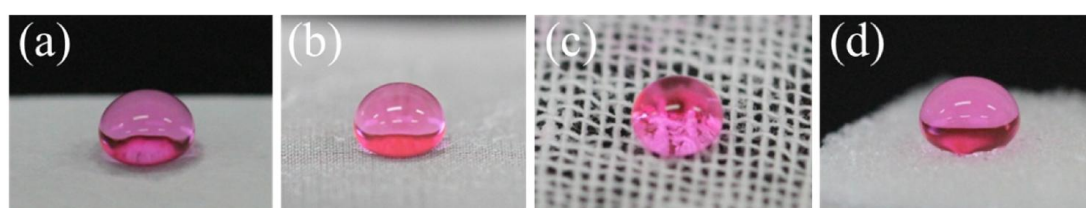


Figure 7. Digital images of a water droplet (dyed with Dil) on various modified materials: filter paper (a), stainless steel mesh (b), gauze (c), and PU foams (d).

3.5 Applications

Recently, more and more research involved in the preparation and applications of the superhydrophobic PP materials was reported.⁴⁰⁻⁴⁵ The prepared PP surfaces with tunable water adhesion also exhibit many potential applications. For example, the surface with high water adhesion can be used as “mechanical hand” to transport small droplet without any loss. A 5 μ L water droplet was placed on A2I1 (low adhesion), and then A6I1 (high adhesion) was lowered down to touch the droplet and then pulled up. The droplet is transferred from A2I1 to A6I1 without any loss (Figure 8a). In addition, the composite coating cannot be swollen or dissolved by many organic solvents, and exhibits superhydrophobic under oil (the left image in Figure 8b). The special microstructures (both the nanostructures on the microspheres and the space between each microsphere) and the superhydrophobicity indicate the applications as the slippery surfaces.⁴⁶⁻⁴⁸ After A4I1 coating was infused with hexadecane, the water droplet can slide off on the slippery surface and finally stick to the blank glass substrate at the bottom (the right image in Figure 8b). PP coatings with low adhesion to substrate like i-PP and A2I1, can be easily transferred to other substrates via transfer printing method. For example, star, clover and bear-like patterns are prepared by double-side tape, with one side pasted on a glass slide, and the other pressed on i-PP coating. Corresponding, i-PP patterns would be peeled off the coating and were stamped on the tape, and the superhydrophobicity remained on the obtained patterns (Figure 8c). These patterned superhydrophobic surfaces can be widely used in water harvesting⁴⁹, cell adhesion⁵⁰, crystal fabrication⁵¹, lab-on-chip devices⁵² and microchannel devices⁵³.

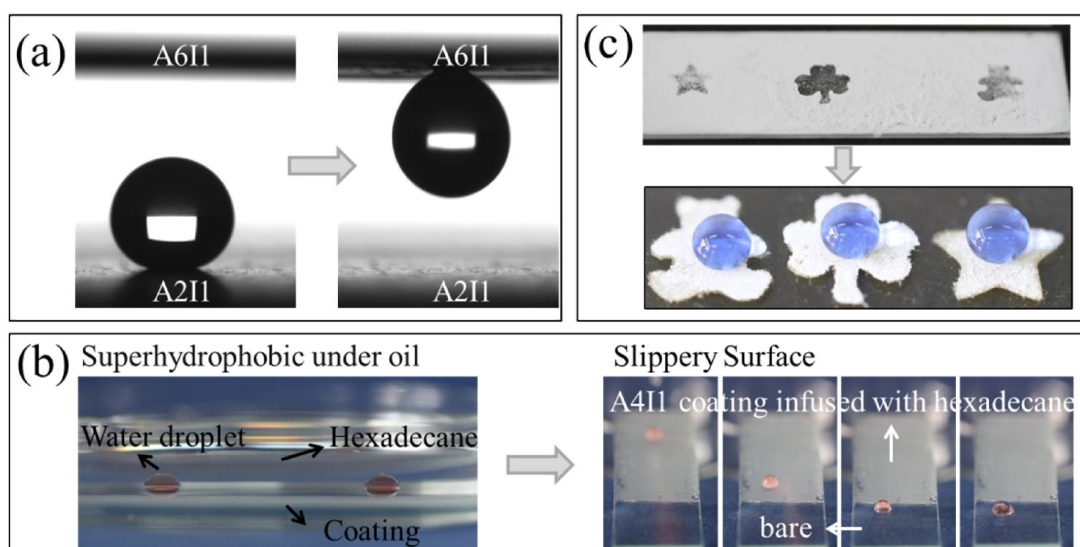


Figure 8. Applications of the PP coatings: no-loss liquid transportation (a), resistance to organic solvent and the superhydrophobicity under oil (the left image), and the utility as slippery surfaces (the right image) (b), and the preparation of superhydrophobic patterns by transfer printing (c).

4. Conclusions

In summary, PP coatings with CA larger than 150° , and the dynamic behaviors tunable from slippery to sticky are achieved via a simple casting method by modulating the mass ratio of a-PP to i-PP in the solution. The distinct crystallization abilities of a-PP and i-PP are the main reason for the formation of various microstructures on the as-prepared composite coatings, thus different wetting states from Cassie to Cassie impregnating wetting state are obtained. In addition, the mechanical properties of the coatings are improved by adding a-PP into i-PP. Furthermore, this feasible method can be utilized to modify various substrates. These ultra water repellent surfaces can be applied in self-cleaning, no-loss liquid

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transportation, slippery materials and patterned superhydrophobic surfaces due to their distinct adhesion behaviors.

Associated Content

Supporting Information

WXRD patterns and DSC results of different PP coatings. SEM images of A4I1 coating before and after etched by cyclohexane. SEM images and CA of smooth i-PP and a-PP films. Low magnification SEM images of the PP coatings. The morphology evolution of i-PP, a-PP and A2I1 coatings investigated by OM. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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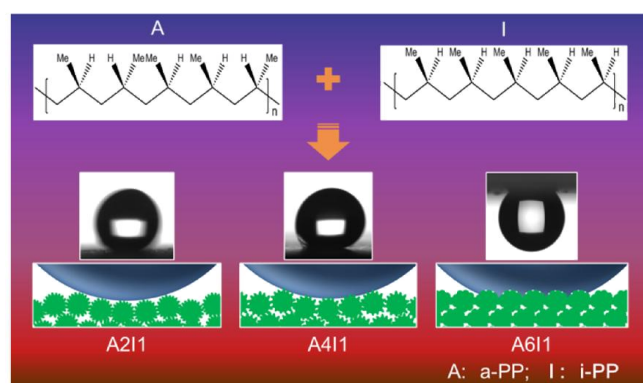
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